Customs Clearance Management Platform — Business Requirements Document

Oct 12, 2026 · Draft v0.1

1. Document control

This is draft v0.1. It turns the client's hand-drawn "Delivery Clearance Process" into requirements the team can review, refine and take into SDLC.

Item	Value
Document	Business Requirements Document (BRD)
Product (working name)	ClearPort: customs clearance and delivery management for port clearing agents
Version	0.1, draft for review
Source material	3 handwritten sketches from the client: the end-to-end flowchart, stage detail sheet 1 (Document Received, Bayan, DO Application), stage detail sheet 2 (Inspection, CCRO, Truck Assignment, Physical Inspection, Container Delivery)
Next review	Walkthrough with the client's operations lead, then sign-off

How to read the requirements

- IDs follow `MODULE-NNN`, for example `JOB-001`. IDs stay stable through refinement so they can be traced into user stories, test cases and releases.
- Priority uses MoSCoW: **M** = Must for launch, **S** = Should, **C** = Could, **W** = Won't for now.
- Phase: **P1** = MVP, **P2** = scale-up, **P3** = platform. Section 6 defines the phases.
- Anything marked *[Confirm]* is an assumption made from the sketches. The client must validate it before design starts. Section 14 lists all of them.

2. Executive summary

We will build a system of record for every Bill of Lading (BL) a clearing agent handles. It tracks each container from the day documents arrive until it is delivered, and it closes the job with every payment and proof attached.

The problem today. The client runs a 9-stage clearance flow in two parallel tracks: the customs declaration (Bayan) and the shipping-line Delivery Order (DO). The tracks join at the CCRO release. After that, work happens container by container: trucks, physical inspection, release and delivery. This is usually coordinated over WhatsApp, phone calls, spreadsheets and paper files. The results are predictable:

- No one can see, in one place, where every BL and container stands.
- Government fees paid by PROs are reconciled late or never billed back to the customer.
- Demurrage and detention charges build up silently when a stage stalls.
- Customers call again and again to ask "where is my container?".
- When one person is absent, the knowledge of where each job stands goes with them.

The product. A web app for the back office and a mobile app for PROs, drivers and field agents. Together they:

1. Capture a BL once, with its containers, and drive it through a configurable stage workflow.
2. Run the Bayan track and the DO track in parallel. Block CCRO until both are complete, together with inspection.
3. Switch to container-level tracking for trucks, physical inspection and delivery. A BL closes automatically once its last container is delivered.
4. Record every amount paid against the BL. Bill it back to the customer, and alert the team before free time runs out.
5. Let customers track their own shipments on a portal and through WhatsApp updates.

Why it wins. Global freight suites such as CargoWise are built for large forwarders. They are expensive, slow to adopt and weak on local stages such as Bayan, PRO runs and CCRO. Most small and mid-size clearing agents therefore still run on spreadsheets. A focused, mobile-first, Arabic/English product built around the exact stages these agents use can be adopted in days, not months. It can then grow into document AI, trucking marketplaces and integrations with customs and port systems.

Business outcomes targeted for the first 12 months after go-live *[Confirm baselines with client]*

- At least 30% shorter average time from document receipt to delivery.
- 100% of PRO-paid amounts captured and billed back.
- A measurable drop in demurrage and detention charges caused by delays on the agent's side.
- 50% fewer status calls from customers.

3. Business context and market opportunity

The best opening is the long tail of small and mid-size licensed clearing agents in GCC ports. Their work is regulated, document-heavy and deadline-driven, yet most of them still run on spreadsheets and messaging apps. *Market sizing in this section is qualitative. A desk study of licensed agent counts and BL volumes per port is a next step (section 15).*

Who the customer is. A customs clearing agent (customs broker) works for importers. It files the customs declaration (Bayan), pays duties and fees through its PROs, gets the DO from the shipping line, and obtains the release. Then it trucks the containers out of the port to the consignee. It earns a service fee per BL or per container, and it often pays government and port charges up front for its customers.

Pain points, ranked by commercial impact

#	Pain point	Business cost	How the product answers it
1	Money paid up front (duty, Bayan fees, DO charges, inspection fees) is tracked on paper	Cash leakage; amounts never billed back; slow collections	Disbursement ledger on every stage; automatic re-billing (section 8.9)
2	Free time expires unnoticed (port storage, line detention)	Demurrage and detention charges, often absorbed by the agent	Free-time clock per container with escalating alerts (section 9)
3	No live view of jobs across staff	Missed hand-offs; a manager has to chase status by phone	Kanban and dashboard by stage; SLA timers (section 8.10)
4	Customers keep asking "where is my container?"	Staff time; poor customer experience	Customer portal and WhatsApp milestone updates
5	Field work (PRO runs, trucks, inspections) is invisible	Delays found late; disputes over driver hand-overs	Mobile app with photo proof, GPS and e-POD
6	Knowledge sits in people's heads	Risk when someone is away; slow onboarding	One BL record with full history and documents

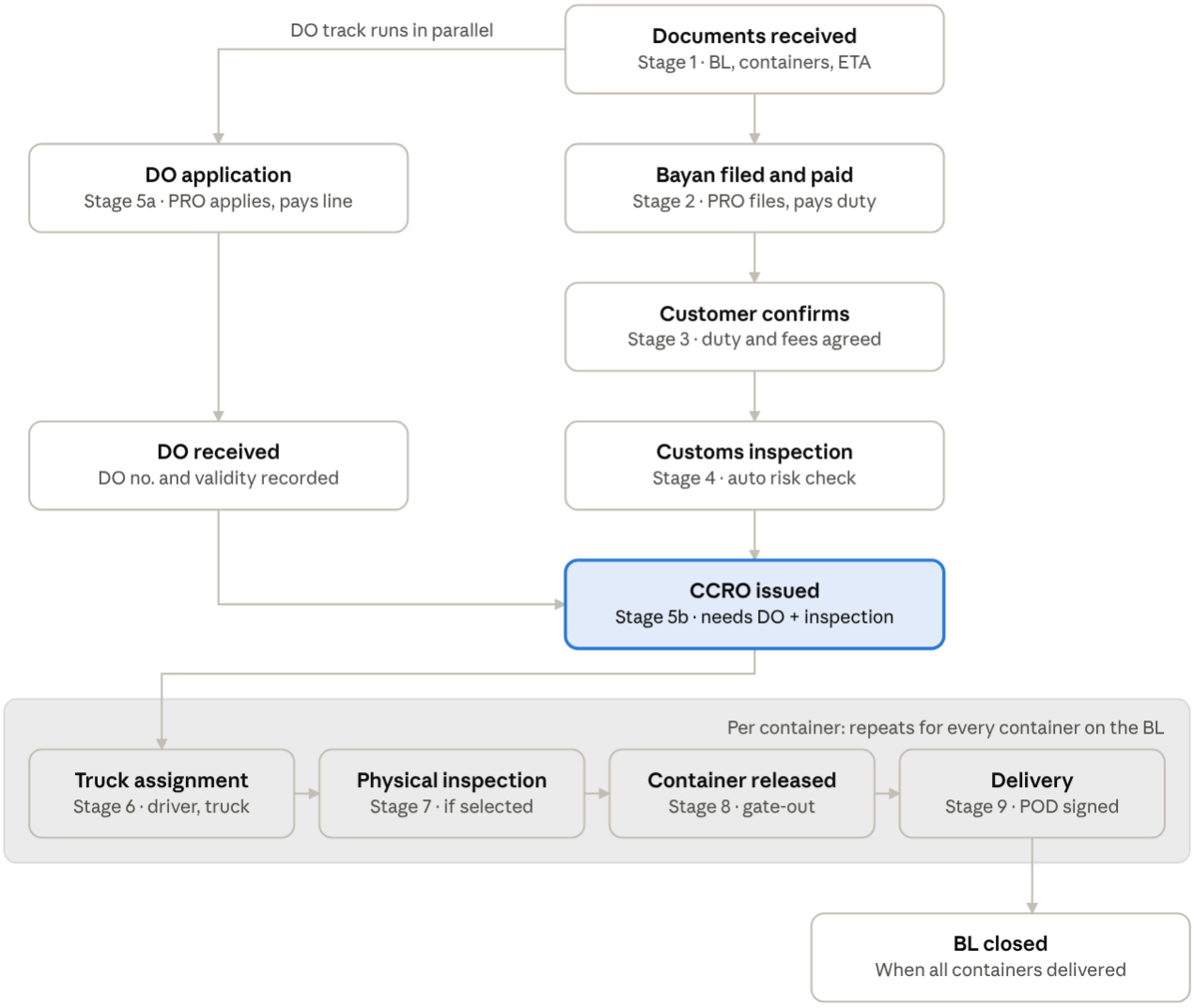
Competitive landscape

Alternative	Strength	Weakness for this segment	Our position
Spreadsheets + WhatsApp (status quo)	Free, familiar	No control, no audit, no billing link	Feels as quick as WhatsApp, with the control of an ERP
Global forwarding suites (CargoWise, Magaya and similar)	Deep functionality, global footprint	High licence and implementation cost; generic stages; months to roll out	Local stages built in; live in a week; per-BL pricing
Local ERPs and accounting tools adapted with custom fields	Accounting is strong	No operational workflow, no field app, no customer visibility	Operations first, with an accounting export
Government and port portals	Authoritative status	One portal per authority; no view across jobs	The agent's single pane over every portal, with integrations later (section 11)

Differentiation in one line: the only clearance tool built around the agent's real stages (Bayan, DO, CCRO, truck, physical inspection), with money and field work captured as they happen.

4. As-is process analysis

The client's flow has two phases. In the first, two tracks run in parallel for the whole BL: the customs track (Bayan, customer confirmation, inspection) and the shipping-line DO track. Both must finish before the CCRO can be obtained. In the second phase, the work splits by container: each container is trucked, inspected if selected, released and delivered. The BL closes after the last delivery.



As-is delivery clearance flow · 9 stages, 2 parallel tracks, 1 gate

CCRO is the single gate where the two tracks meet. It is the step most likely to stall, and the step customers care about most.

What each sketch says and where it lands in this BRD

Sketch element	What we read from it	Requirements
Flowchart: Customer → Document Received	Customer sends documents; the job starts	JOB-001 to JOB-005
Flowchart: Bayan ↔ PRO, "Paid Amount"	A PRO handles the declaration and pays fees that must be tracked	BAY-001 to BAY-003, FIN-001, FIN-002

Sketch element	What we read from it	Requirements
Flowchart: "Communicate customer" → "Done"	Duty and charges are confirmed with the customer before moving on	BAY-004, BAY-005
Flowchart: Document Received → DO Apply → DO Received	DO runs in parallel with Bayan	DO-001 to DO-004
Flowchart: Inspection, with a "?" triangle	Inspection outcome is a decision point *[Confirm meaning]*	INS-001 to INS-003
Flowchart + sheet 2: CCRO needs PRO, DO, inspection completion	Hard gate with three conditions	CCR-001 to CCR-003
Sheet 1: #BL with customer, container list with unique serial numbers, BL no., ETA, commodity, shipping details (optional), port details	BL header and container list data model	Section 7, JOB-001, JOB-002
Sheet 1: Bayan and DO Application each with PRO, status update, remarks	Every stage has an assignee, a status history and remarks	Section 7 design rules, OPS-003
Sheet 2: Inspection "(Auto process)" with status and comments	Result comes from customs, recorded by staff	INS-001
Sheet 2: Truck Assignment " (container based process)" with driver, vehicle, container, physical inspection agent, delivery details	Work moves to container level from here	TRK-001 to TRK-004
Sheet 2: Physical Inspection with containers and status	Result per container	PHY-001 to PHY-003
Sheet 2: Container delivery → "BL process closed"	Delivery details per container; auto-close of the BL	DEL-001 to DEL-005

Gaps the sketches do not cover (we added these as Should/Could items):

- Free time, demurrage and detention.
- Customer billing and invoicing.

- Empty container return.
- Exception handling: a container held, a query on the Bayan, an expired DO.
- Customer self-service tracking.

5. Stakeholders, personas and roles

Seven roles touch a BL. Three of them work mostly in the field, so mobile is a launch requirement, not a later add-on.

Role	Who	Main jobs in the product	Channel	Phase
Agency owner / manager	Owner or general manager of the clearing agency	Sees every job, delay and amount outstanding; approves write-offs; reviews profit per customer	Web dashboard, mobile summary	P1
Operations coordinator (documentation)	Back-office staff who open and drive jobs	Creates BLs, uploads documents, moves stages, talks to customers, assigns PROs and trucks	Web	P1
PRO (Public Relations Officer)	Staff who deal with customs, the port and shipping lines in person or online	Files Bayan and DO applications, records amounts paid with receipts, updates status from the field	Mobile first	P1
Physical inspection agent	Staff or contractor present at the customs inspection	Records inspection result per container with photos and remarks	Mobile	P1

Role	Who	Main jobs in the product	Channel	Phase
Driver / transporter	In-house driver or a third-party trucking company	Accepts the trip, confirms pick-up, gate-out and delivery with proof of delivery (POD)	Mobile (light app or a WhatsApp link)	P1 basic, P2 full
Accounts	Finance staff	Reconciles PRO payments, raises invoices, tracks receivables	Web	P1 ledger, P2 invoicing
Customer (importer / consignee)	Client of the agency	Sends documents, approves charges, tracks shipments, downloads invoices and POD	Customer portal, WhatsApp, email	P1 notifications, P2 portal
Platform super-admin	Our company (SaaS operator)	Onboards agencies (tenants), manages subscriptions, gives support	Admin console	P2

Access principles

- Role-based access control (RBAC). Permissions are set per stage, so a PRO can update Bayan but cannot close a BL.
- Access can be limited by branch or port where the agency has several offices.
- Customers see only their own BLs, and only fields marked as customer-visible.

6. Scope and phased roadmap

The MVP digitises the client's 9-stage flow exactly as drawn, including the money paid at each stage. Integrations, AI and marketplace features come after we have proven the client uses it every day.

In scope

- Import clearance of containerised sea cargo (FCL), from document receipt to delivery and BL closure.
- Both tracks shown in the sketch: Bayan with inspection, and the shipping-line DO. They join at CCRO.
- Container-level execution: truck assignment, physical inspection, release and delivery.
- Amounts paid per stage (disbursements), customer charges, invoicing and receivables.
- Customer communication: notifications, approvals and a tracking portal.
- Multiple agencies (tenants) on one platform, so the product can be sold beyond the first client.

Out of scope for now *(candidates for later phases)*

- Export clearance, transit and re-export. The data model must still allow them later.
- LCL / consolidated cargo, air cargo and land border clearance.
- A full general-ledger accounting system. We export to the client's accounting tool instead.
- Booking freight, and buying or selling ocean freight.
- Fleet maintenance and driver payroll.

Phases

Phase	Goal	Key capabilities	Exit gate
P1 — MVP (pilot with the client)	Replace spreadsheets and WhatsApp for the core flow	BL and container intake; the 9-stage workflow with parallel tracks; PRO mobile updates with receipts; truck assignment; physical inspection and delivery with POD; disbursement ledger; dashboard and kanban; WhatsApp/email notifications; audit trail; English and Arabic	Every new BL at the client runs in the system for 4 consecutive weeks; less than 5% of updates made outside the system

Phase	Goal	Key capabilities	Exit gate
P2 — Scale and monetise	Make it saleable to other agents and directly profit-driving	Free-time and demurrage engine; invoicing and re-billing; customer portal; tariff and quotation engine; document AI (OCR of BL/invoice); driver app with GPS; SLA analytics; tenant self-onboarding and subscriptions	5 or more paying agencies; customer portal used on 50% of BLs
P3 — Platform	Network effects and deeper integrations	Integrations with customs, port and shipping-line systems; a trucking marketplace; customer API; predicted clearance time; WhatsApp chatbot; exports and transit; payment gateway	Integration partners live; marketplace transactions

7. Core domain model

The BL is the job, and the container is the unit of execution. Stages 1–5 run once per BL. Stages 6–9 run once per container. The BL closes only when every container is delivered. This mirrors the client's note that truck assignment is a "container based process".

Main entities

Entity	Key attributes (from the sketches plus what clearance needs)	Relationships
Customer	Name, trade licence, importer code, contacts, credit terms, preferred channel, tariff	Has many BLs

Entity	Key attributes (from the sketches plus what clearance needs)	Relationships
BL (Clearance job)	Job no. (auto), BL no. (unique per tenant), customer, vessel / voyage, shipping line, port of loading, port of discharge, ETA / ATA, commodity, HS code(s), shipping details (optional), assigned coordinator, overall status, priority	Has many containers, stages, documents, payments, comments
Container	Container no. (ISO 6346, check digit validated), size/type (20GP, 40HC, reefer…), seal no., gross weight, packages, free-time expiry, current status, current location	Belongs to a BL; has one or more truck trips, a physical inspection result, a delivery
Stage instance	Stage type, status, assignee (PRO / agent), start and finish times, SLA due, remarks, reference no. (Bayan no., DO no., CCRO no.)	Belongs to a BL or a container
Payment / disbursement	Stage, fee type, amount, currency, paid by (PRO), paid on, receipt image, billable to customer (Y/N), markup	Belongs to a stage; rolls up to the BL invoice
Document	Type (BL, commercial invoice, packing list, certificate of origin, Bayan copy, DO, CCRO, POD…), file, version, uploaded by, customer-visible flag	Belongs to a BL or a container
Truck trip	Transporter, driver (name, mobile, licence), vehicle (plate no., type), container, physical inspection agent, pick-up and delivery address, times, POD	Belongs to a container
Comment / activity	Author, text, mentions, internal or customer-visible, timestamp	Belongs to any entity; feeds the audit trail

Stage catalogue and gates

#	Stage	Level	Can start when	Statuses	Captures
1	Document Received	BL	Customer sends documents	Draft → Received → Verified	BL data, container list, ETA, commodity, port details, documents
2	Bayan (customs declaration)	BL	Stage 1 verified	Not started → Filed → Query raised → Approved → Duties paid	PRO, Bayan no., amount paid, status updates, remarks
3	Customer communication	BL	Bayan approved or duty assessed	Pending → Sent → Confirmed	Duty and charges shared, customer confirmation ("Done" in sketch)
4	Inspection (customs, auto)	BL	Stage 3 confirmed	Awaiting → Cleared (no inspection) / Inspection required → Completed	Result, comments; sets which containers need physical inspection *[Confirm]*
5a	DO Application	BL	Stage 1 verified (runs in parallel with stages 2–4)	Not started → Applied → Charges paid → DO received	PRO, DO no., line charges, validity date, remarks
5b	CCRO	BL	DO received AND inspection completed AND PRO assigned	Not started → Applied → Issued	CCRO no., date, amount paid
6	Truck Assignment	Container	CCRO issued	Unassigned → Assigned → At port	Driver, vehicle, container, inspection agent, delivery details
7	Physical Inspection	Container	Truck at port, if required	Not required / Scheduled → Passed / Held	Status, remarks, photos
8	Container Released	Container	Inspection passed or not required	Released → Gated out	Gate-out time, release reference
9	Delivery	Container	Gated out	In transit → Delivered → Empty returned	Delivery details, POD (photo + signature), empty return time
—	BL Closure	BL	All containers delivered AND all payments reconciled	Open → Closed	Close date, final invoice no.

Design rules for the SDLC team

- The stage catalogue, its gates and its statuses are **configuration, not code**. Each agency, port or country can then adapt the flow without a release.
- Every status change is an event with who, when, from → to, and remarks. Timelines, SLAs, notifications and the audit trail all build on these events.
- Containers within one BL may move at different speeds. The BL status is derived from the slowest container, for example "3 of 5 delivered".

8. Functional requirements

There are 13 modules. Every Must-have for P1 traces back to a box in the client's sketches. Should and Could items are where we add value beyond the sketch.

8.1 Customer management (CUS)

ID	Requirement	Pri	Phase
CUS-001	Create and maintain customers: name, trade licence no., importer/customs code, address, several contacts with roles	M	P1
CUS-002	Store each customer's preferred notification channel (WhatsApp, email, SMS) and language (English/Arabic)	M	P1
CUS-003	Store credit terms, credit limit and an agreed tariff per customer	S	P2
CUS-004	Customer 360 view: open BLs, containers in progress, amount outstanding, average clearance time	S	P2
CUS-005	Warn when a new BL is opened for a customer over their credit limit	C	P2

8.2 Job intake — Document Received (JOB)

ID	Requirement	Pri	Phase
JOB-001	Create a clearance job keyed by BL no. Fields: customer, BL no., ETA, commodity, shipping details (optional), port details (port of loading, port of discharge, terminal)	M	P1

ID	Requirement	Pri	Phase
JOB-002	Add a list of containers with unique numbers ("Container 1 … Container N" in the sketch). Allow bulk paste or CSV import, and validate the ISO 6346 check digit	M	P1
JOB-003	Block duplicate BL numbers per tenant, and the same container on two open jobs	M	P1
JOB-004	Upload the documents received (BL, commercial invoice, packing list, certificate of origin, others) and run a completeness checklist per commodity	M	P1
JOB-005	Auto-generate a job number; assign a coordinator and a priority	M	P1
JOB-006	Receive documents by email-in: forwarding to a job mailbox creates a draft job with attachments	S	P2
JOB-007	Document AI: read the BL and invoice to pre-fill BL no., containers, ETA, commodity and HS codes for the user to confirm	S	P2
JOB-008	Pull the vessel ETA from a vessel or line tracking feed and update it automatically	C	P3

8.3 Bayan — customs declaration (BAY)

ID	Requirement	Pri	Phase
BAY-001	Assign a PRO to the Bayan stage; the PRO gets a mobile task	M	P1
BAY-002	Record the Bayan reference no., declaration date and status updates (Filed, Query raised, Approved, Duties paid) with remarks on each	M	P1
BAY-003	Record the amount paid (duty, VAT, fees) with a photo of the receipt; flag each as billable to the customer	M	P1
BAY-004	When duties are assessed, prompt the coordinator to tell the customer ("Communicate customer" in the sketch) and record the customer's confirmation ("Done")	M	P1

ID	Requirement	Pri	Phase
BAY-005	Customer approves duty and charges through a one-tap link (WhatsApp/email) instead of a phone call	S	P2
BAY-006	Draft the declaration lines (HS code, value, origin) from the invoice so the PRO files faster	C	P3
BAY-007	Integrate with the customs system to submit and get status automatically	C	P3

8.4 Delivery Order — DO (DO)

ID	Requirement	Pri	Phase
DO-001	Start a DO application as soon as documents are verified, running in parallel with Bayan	M	P1
DO-002	Assign a PRO; record status updates (Applied, Charges paid, DO received) and remarks	M	P1
DO-003	Record DO no., the shipping line's charges paid (with receipt) and the DO validity date; upload the DO copy	M	P1
DO-004	Alert before the DO validity date expires when containers are still undelivered	S	P1
DO-005	Keep a directory of shipping lines with their contacts and standard charges	S	P2

8.5 Inspection and CCRO (INS, CCR)

ID	Requirement	Pri	Phase
INS-001	Record the customs inspection result ("auto process" in the sketch): status and comments	M	P1
INS-002	Mark which containers are selected for physical inspection, so that stage 7 applies only to those	M	P1 *[Confirm]*
INS-003	Log fees for inspection, scanning and examination as payments	M	P1

ID	Requirement	Pri	Phase
CCR-001	Lock CCRO until all three gates from the sketch are met: PRO assigned, DO received, inspection completed. Show which gate is missing	M	P1
CCR-002	Record the CCRO application, CCRO no., date issued, amount paid and a document copy	M	P1
CCR-003	Notify the coordinator and the customer when the CCRO is issued, the moment the cargo is cleared	M	P1

8.6 Truck assignment (TRK) — per container

ID	Requirement	Pri	Phase
TRK-001	Per container, assign driver details, vehicle details, the physical inspection agent and delivery details (address, contact, time window)	M	P1
TRK-002	Assign several containers at once to trucks or a transporter	M	P1
TRK-003	Keep master data for drivers, vehicles and third-party transporters, including driver licence and vehicle registration expiry	M	P1
TRK-004	Send the driver a trip link (no install needed) with the container no., pick-up point, delivery address and a status button	S	P1
TRK-005	Record trucking cost per trip (own fleet or transporter rate) for job profitability	S	P2
TRK-006	Live GPS tracking of trips and an ETA for the customer	C	P2
TRK-007	Trucking marketplace: post containers for transporters to bid on or accept	C	P3

8.7 Physical inspection and release (PHY, REL) — per container

ID	Requirement	Pri	Phase
PHY-001	The inspection agent records the result per container (Passed / Held / Partial) with remarks from mobile	M	P1
PHY-002	Attach photos and videos with timestamp and geotag as evidence	S	P1
PHY-003	A held container raises an exception task with a reason, an owner and a due date; the rest of the BL can continue	M	P1
REL-001	Mark the container released / gated out with time and reference	M	P1

8.8 Delivery and BL closure (DEL)

ID	Requirement	Pri	Phase
DEL-001	Record delivery details per container: delivered to, date and time, receiver name	M	P1
DEL-002	Electronic POD: receiver signature on the driver's phone plus a photo; a PDF is generated and shared with the customer	S	P1
DEL-003	Track empty container return to the line's depot and the return deadline (detention)	S	P2
DEL-004	Auto-close the BL ("BL process closed" in the sketch) when all containers are delivered and every payment is reconciled; otherwise list what blocks closure	M	P1
DEL-005	Reopen a closed BL only with a manager role and a reason	M	P1

8.9 Disbursements, billing and receivables (FIN)

ID	Requirement	Pri	Phase
FIN-001	Ledger per BL of every amount paid at any stage: who paid, how, receipt, billable or not	M	P1
FIN-002	PRO cash advance and settlement: issue a float to a PRO, then reconcile the receipts against it	S	P1
FIN-003	Generate the customer invoice from the service fee (per BL or per container from the tariff) plus billable disbursements, with VAT	S	P2
FIN-004	Ageing of receivables and payment reminders to the customer	S	P2
FIN-005	Profit per job (fees minus costs) and per customer	S	P2
FIN-006	Export to accounting (Excel/CSV first, then connectors such as Zoho Books, QuickBooks, Tally)	S	P2
FIN-007	Quotation engine: estimate total landed clearance cost for a prospect from the tariff	C	P2
FIN-008	Online payment link on invoices	C	P3

8.10 Operations dashboard and tracking (OPS)

ID	Requirement	Pri	Phase
OPS-001	Kanban board of BLs by stage, with filters for customer, PRO, port, ETA and priority	M	P1
OPS-002	Container board showing every container's current stage and location	M	P1
OPS-003	BL timeline: every stage, status change, payment, document and comment in order	M	P1
OPS-004	SLA timers per stage (configurable targets). Stuck jobs turn amber, then red	S	P1
OPS-005	"My tasks" list per user, across web and mobile	M	P1
OPS-006	Global search by BL no., container no., Bayan no., DO no., or customer	M	P1

8.11 Notifications and customer experience (NTF, CXP)

ID	Requirement	Pri	Phase
NTF-001	In-app and email notifications for assignments, gate unlocks, SLA breaches and mentions	M	P1
NTF-002	WhatsApp Business milestone messages to the customer: documents received, duty to approve, cleared, out for delivery, delivered (with POD)	S	P1
NTF-003	Each tenant chooses which milestones go to customers, and edits the templates in English and Arabic	S	P2
CXP-001	Customer portal: list of BLs and containers with live status, documents, invoices and POD downloads	S	P2
CXP-002	Customer uploads documents and creates a new clearance request from the portal	S	P2
CXP-003	WhatsApp chatbot: the customer sends a BL or container no. and gets the status back	C	P3

8.12 Documents (DOC)

ID	Requirement	Pri	Phase
DOC-001	Store documents per BL and container with type, version history and preview	M	P1
DOC-002	Capture from the mobile camera with auto-crop, saved as PDF	S	P1
DOC-003	Mark each document internal or customer-visible	M	P1
DOC-004	Retention policy and archive per tenant, in line with the customs record-keeping period *[Confirm period]*	S	P2

8.13 Administration, configuration and audit (ADM)

ID	Requirement	Pri	Phase
ADM-001	Users, roles and permissions per stage and action; branch/port scoping	M	P1
ADM-002	Configure the stage catalogue: stages, statuses, gates, required fields, SLA targets	S	P1
ADM-003	Master data: ports, terminals, shipping lines, fee types, container types, document types	M	P1
ADM-004	Immutable audit log of every create, update, status change and payment	M	P1
ADM-005	Multi-tenant onboarding, subscription plans and usage metering per BL/container	S	P2
ADM-006	Bilingual UI (English/Arabic, right-to-left) and multi-currency	M	P1

9. High-potential differentiating features

Eight features turn this from a workflow tracker into a product that agents pay for. Each one either saves money the agent loses today, or makes the agent look better to its own customers.

Feature	What it does	Why it sells	Phase
Free-time and demurrage guard	Holds the port free-time and line detention free days per container, counts down from discharge, and escalates at T-3, T-1 and expiry. It shows the projected charge if nothing moves today.	Cuts demurrage and detention, often the single largest avoidable cost in clearance. Easy to prove ROI in a sales demo.	P2 (basic expiry alert in P1)

Feature	What it does	Why it sells	Phase
Zero-leakage disbursement capture	Every PRO payment is logged on mobile with a receipt photo at the moment of payment, then flows straight to the customer invoice.	Recovers money that slips through on paper today; pays for the subscription by itself.	P1
Parallel-track gatekeeper	Shows Bayan and DO side by side and explains exactly what blocks CCRO.	Removes the most common hand-off failure in the client's flow.	P1
Document AI intake	OCR plus extraction from BL, invoice and packing list. It pre-fills the job, containers and HS codes, and flags mismatches (for example container count on BL vs. packing list).	Cuts job creation from about 15 minutes to under 2 *[estimate, to validate in pilot]*; fewer filing errors and penalties.	P2
Customer self-service tracking	Branded portal and WhatsApp milestones with POD attached.	Agents can sell transparency to their importers; fewer status calls.	P1 WhatsApp, P2 portal
Field-first mobile app	PRO, inspection agent and driver flows designed for one thumb, with offline queueing for port areas with poor signal.	Field updates arrive in real time instead of at the end of the day.	P1
Clearance-time intelligence	Learns typical stage durations by port, commodity and line, and predicts the delivery date per container.	Sales-grade promise to customers; spots bottlenecks for the manager.	P3

Feature	What it does	Why it sells	Phase
Trucking marketplace	Lets an agent post containers to a network of vetted transporters who accept or bid.	Network effect and a transaction revenue stream beyond subscriptions.	P3

10. Non-functional requirements

The product handles customs and financial records for many agencies. Tenant isolation, an audit trail and mobile reliability are the non-negotiables.

ID	Area	Requirement
NFR-001	Multi-tenancy	Every record is scoped to a tenant; no query can return another tenant's data. Tenant ID is enforced at the data-access layer.
NFR-002	Security	HTTPS everywhere; passwords hashed; optional 2FA (mandatory for owner and accounts roles); session timeout; OWASP Top 10 covered in testing.
NFR-003	Data residency and privacy	Host in-region where the client's regulator or customers require it *[Confirm country]*; comply with the applicable data protection law.
NFR-004	Audit	Status changes, payments and edits are append-only and kept for at least the customs record-keeping period.
NFR-005	Performance	Dashboard and search respond in under 2 s at the 95th percentile with 50,000 BLs per tenant; mobile actions confirm in under 1 s locally.
NFR-006	Availability	99.5% monthly uptime in P1, 99.9% from P2; daily backups with 30-day retention; recovery point under 24 h and recovery time under 4 h.
NFR-007	Mobile and offline	Android and iOS; field actions and photos queue offline and sync when connected; works on mid-range Android phones.
NFR-008	Localisation	English and Arabic with full right-to-left layout; local date formats; multi-currency with a base currency per tenant.

ID	Area	Requirement
NFR-009	Usability	A new coordinator can create and progress a BL without training in under 10 minutes; all field flows are 3 taps or fewer per update.
NFR-010	Scalability	Horizontal scaling of the API; file storage on S3-compatible object storage.
NFR-011	Observability	Central logging, error tracking and uptime alerts; per-tenant usage metrics.
NFR-012	Configurability	Stage catalogue, fee types, notification templates and SLAs changeable by an admin without a code release.

11. Integrations

P1 needs only messaging, storage and maps. Integrations with government, port and line systems are the long-term moat, but they depend on API access that varies by country, so they sit in P3.

System	Purpose	Direction	Phase	Dependency
WhatsApp Business API	Customer milestones, driver trip links, approvals	Outbound + replies	P1	Meta business verification; message template approval
Email (SMTP + inbound parsing)	Notifications; email-in document intake	Both	P1 / P2	Tenant domain set-up
SMS gateway	Fallback for drivers without WhatsApp	Outbound	P1	Local SMS provider
Object storage	Documents, receipts, photos, POD	—	P1	S3-compatible storage
Maps and geolocation	Delivery addresses, trip GPS	Inbound	P1 / P2	Map provider licence
OCR / document AI service	Extract data from BL, invoice and packing list	Inbound	P2	Model choice; accuracy benchmark on client samples

System	Purpose	Direction	Phase	Dependency
Accounting systems (Zoho Books, QuickBooks, Tally, Excel)	Invoices and payments	Outbound	P2	Client's accounting tool *[Confirm]*
Payment gateway	Customers pay invoices online	Inbound	P3	Local acquiring bank
Customs system (Bayan)	Submit declarations, read status and duty	Both	P3	Authority API access and agent credentials
Port community / terminal system	Container availability, holds, gate-out events	Inbound	P3	Port API or EDI access
Shipping line portals / EDI	DO status, free time, empty return depot	Inbound	P3	Per-line agreements
Vessel tracking (AIS)	Live ETA	Inbound	P3	Data provider subscription

12. Reports, KPIs and success metrics

The north-star metric is **average document-to-delivery time per container**. Every stage event feeds it, and it is what an agent's customers care about.

Operational reports (in product)

Report	Audience	Key measures	Phase
Live operations board	Coordinator, manager	BLs and containers by stage; overdue items; today's deliveries	P1
Stage cycle-time report	Manager	Average and 90th-percentile duration per stage, by port, line, commodity, PRO	P1
Disbursement report	Accounts, manager	Paid vs. billed vs. collected per BL and customer; unreconciled PRO floats	P1
Free-time exposure	Manager	Containers inside 3 days of free-time expiry; projected charges	P2

Report	Audience	Key measures	Phase
Customer statement and ageing	Accounts, customer	Invoices, payments, overdue buckets	P2
Profitability	Owner	Revenue, cost and margin per BL, customer, month	P2
Team productivity	Manager	BLs handled and on-time rate per coordinator / PRO / driver	P2

Product success metrics (how we judge the pilot and the business)

Metric	Target	When
Share of the client's new BLs created in the system	100%	By week 4 of pilot
Field updates made from mobile (vs. back-office on behalf)	⩾ 80%	By week 8
PRO payments with a receipt attached	⩾ 95%	By week 8
Average document-to-delivery time	− 30% vs. baseline	6 months after go-live
Customer status calls	− 50% vs. baseline	6 months after go-live
Paying agencies	5 or more	End of P2
Monthly logo churn	< 2%	From P2

13. Commercial model

We recommend a SaaS subscription by tier, with a usage component per BL. Price then grows with the agent's volume, and the first client becomes the design partner and reference customer. *Prices are placeholders until pricing research is done.*

Tier	For	Includes	Pricing basis
Starter	Small agents, up to about 50 BLs per month	Core workflow, mobile app, disbursement ledger, email notifications, 5 users	Flat monthly fee

Tier	For	Includes	Pricing basis
Professional	Growing agents	Starter + WhatsApp, customer portal, invoicing, free-time guard, document AI quota, unlimited field users	Monthly fee + per BL above the included volume
Enterprise	Multi-branch agents and forwarders	Professional + integrations, custom workflows, SSO, dedicated support, data residency options	Annual contract

Other revenue lines

- One-off onboarding and data migration fee.
- Document AI pages beyond the quota.
- WhatsApp conversations passed through at cost plus margin.
- P3: commission on trucking marketplace jobs and on payments processed.

Go-to-market

1. Pilot with the client as design partner, at a discounted rate in exchange for a case study.
2. Reach agents through customs broker associations and port communities, and through shipping-line and transporter referrals.
3. Run a demo built around ROI: demurrage avoided plus disbursements recovered, against the subscription cost.

14. Assumptions, risks and open questions

The sketches leave twelve points open. The first four change the data model, so we need them answered before design starts.

Open questions for the client walkthrough

- [ ] Which country and port(s) does the client operate in? Is "Bayan" the national customs system there? This drives integrations, data residency and terminology.
- [ ] What does CCRO stand for in the client's process, and who issues it (customs, the port or the terminal)?
- [ ] Is "Inspection (auto process)" the customs risk-engine result (no inspection vs. inspection required)? Does it select specific containers for physical inspection, or the whole BL?

- [ ] Is a BL ever split across deliveries or customers? Can one job hold several BLs (e.g. a master BL and house BLs)?
- [ ] On the flowchart, a struck-through link runs from Bayan towards DO Apply. Must DO wait for anything from Bayan, or are the two fully parallel?
- [ ] What does the triangle with "?" beside Inspection mean: a decision point, a risk, or a missing step?
- [ ] Physical inspection is drawn after truck assignment. Is the container trucked to an inspection yard first, and is the "physical inspection agent" a staff member or a customs officer?
- [ ] Which fees are paid at each stage, by whom (PRO cash, company card, online), and which are re-billed with markup?
- [ ] Are trucks in-house, third-party, or both? How are drivers paid?
- [ ] What are the typical monthly volumes (BLs, containers) and current stage durations? We need these for KPI baselines.
- [ ] Which accounting tool is used today?
- [ ] Does the client also handle exports, LCL or air cargo that should be in a near-term phase?

Assumptions

#	Assumption	Impact if wrong
A1	Imports of FCL sea cargo only in P1	Data model must widen for LCL / air sooner
A2	Bayan and DO tracks run in parallel; CCRO needs both plus inspection	Gate rules change (configurable, so low impact)
A3	Government and port portals stay manual in P1; staff update status in our app	Double entry until P3 integrations
A4	The client's staff have smartphones with data	Need a lighter SMS / web-link flow

Risks

Risk	Likelihood	Impact	Mitigation
Field staff keep using WhatsApp instead of the app	High	High	3-tap mobile flows; WhatsApp-link trip updates; manager dashboard shows who updates late

Risk	Likelihood	Impact	Mitigation
Government / port APIs unavailable or slow to approve	High	Medium	Product works fully without them; integrations are add-ons
Workflow differs by port or country	Medium	High	Configurable stage catalogue from P1 (ADM-002)
Low document AI accuracy on poor scans	Medium	Medium	Human confirmation step; benchmark on client samples before P2
Building P2/P3 features before the core is adopted	Medium	High	Exit gates per phase (section 6)
Data security incident across tenants	Low	Very high	Tenant isolation tests, penetration test before P2 launch

15. Glossary and next steps

Term	Meaning in this document
BL (Bill of Lading)	The shipping document for the cargo; here, the key of a clearance job
Bayan	The customs declaration filed with customs (name of the declaration and system in several GCC countries) *[Confirm local usage]*
DO (Delivery Order)	Document from the shipping line that authorises release of containers to the consignee or agent
CCRO	Customs/container release order needed before containers leave the port; issued after DO and inspection *[Confirm expansion]*
PRO	Public Relations Officer: staff who deal with government and port bodies and pay fees on the agent's behalf
Physical inspection	Customs examination of the cargo inside a container
POD	Proof of delivery: receiver's signature and photo

Term	Meaning in this document
Demurrage / detention	Charges for keeping containers in the port (demurrage) or outside it (detention) beyond the free time
FCL / LCL	Full container load / less than container load
Disbursement	An amount the agent pays on the customer's behalf and bills back

Next steps toward SDLC

1. Walk through this BRD with the client's operations lead and answer the open questions in section 14.
2. Collect sample documents (5 to 10 real BL packs) and one month of volume and timing data for baselines.
3. Desk study on market size: licensed agents and BL volumes in the target ports.
4. Turn the P1 Must-have requirements into epics and user stories with acceptance criteria. The IDs carry over.
5. Produce wireframes for the four core screens (BL detail, kanban, PRO mobile task, driver trip) and validate them with field staff.
6. Write the technical solution design: architecture, data model, tenancy, configurable workflow engine.
7. Agree the pilot plan, success metrics and sign-off, then begin sprint 0.